

Walmart Store Sales — Demand Forecasting

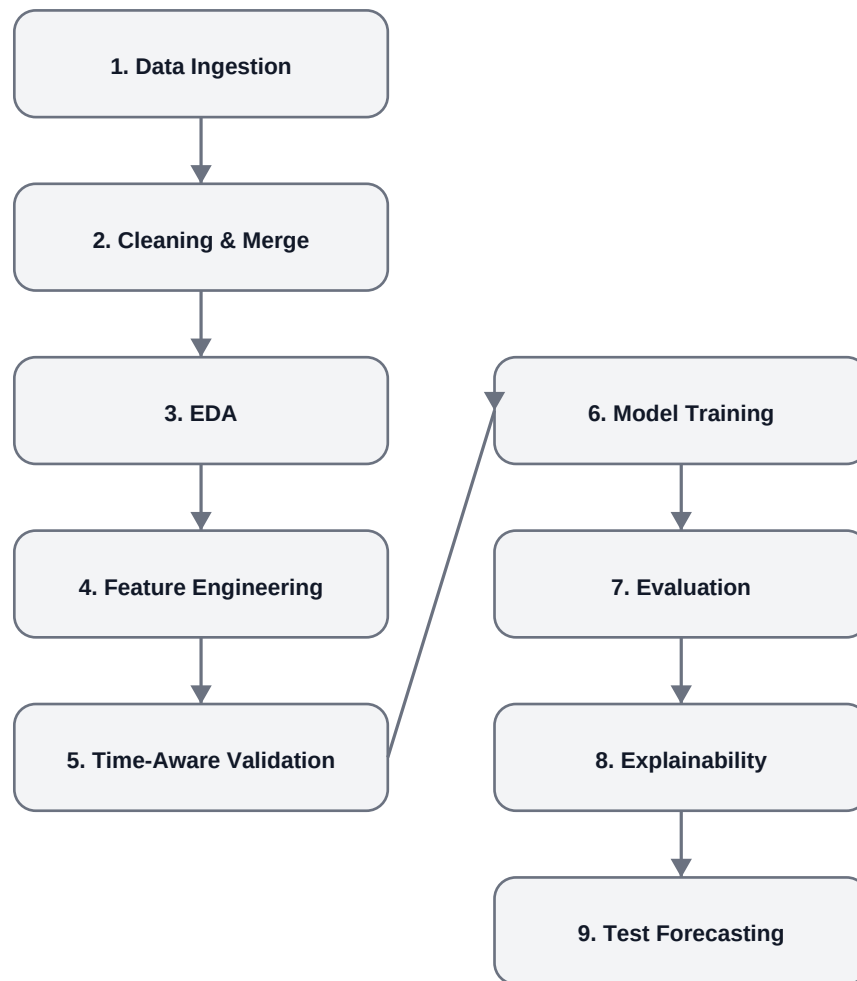
End-to-End Process Flow & Technical Blueprint

AI Engineer Technical Assignment

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Technical blueprint for reproducing the executed **demand_forecasting_v2(1).ipynb** pipeline, from raw data ingestion through validation, feature engineering, model comparison, explainability, and test forecasting.

1. End-to-End Process Flow



Pipeline: raw CSVs → merge and validation → cleaning → EDA → calendar/lag/rolling features → chronological hold-out + expanding-window CV for tuning → Linear Regression / Random Forest / tuned LightGBM → evaluation → explainability → recursive test forecasting → Kaggle submission CSV.

2. Data Ingestion

The notebook loads stores.csv, train.csv, test.csv and features.csv. Recorded shapes are 45 stores, 207,323 training rows, 115,064 test rows and 8,190 feature rows. Store metadata are merged on Store; weekly external features are merged on Store + Date. The duplicate IsHoliday field from features.csv is removed before merging.

Technical rationale: This creates the modeling table containing historical sales, store attributes, external economic signals, markdowns and holidays.

3. Data Cleaning & Transformation

MarkDown1–5 are filled with zero. CPI and Unemployment are forward/backward filled within Store because they are slow-moving signals. The notebook reports 607 negative Weekly_Sales rows (0.29%) and retains them. Store Type is one-hot encoded with drop_first=True.

Technical rationale: Rows without sufficient 52-week history are excluded from the primary modeling table so lag_52 is available.

4. Exploratory Data Analysis

EDA covers total sales over time, store-type distributions, holiday impact, correlations, target distribution and department-level analysis. Average sales are about \$17,588 on non-holiday weeks and \$18,994 on holiday weeks, an observed uplift of 8.0%.

Technical rationale: The EDA establishes seasonality, store heterogeneity, holiday effects and department-level demand differences.

5. Feature Engineering

Calendar features: Year, Week, Month, Days_to_Holiday and Near_Holiday. Historical demand features are created per Store + Department: Sales_lag_1, Sales_lag_2, Sales_lag_4, Sales_lag_52 and rolling means over 4, 8 and 12 weeks. External features include store, department, size, temperature, fuel price, CPI, unemployment, markdowns and Store Type.

Technical rationale: The 52-week lag captures annual seasonality; short lags capture recent momentum; rolling means smooth short-term noise; holiday proximity provides an explicit seasonal timing signal.

6. Model Training & Validation

The primary chronological split ends training on 2012-07-20 and begins validation on 2012-07-27. LightGBM tuning uses 5-fold expanding-window TimeSeriesSplit. The winning parameters are num_leaves=63, n_estimators=200, max_depth=6, learning_rate=0.03, colsample_bytree=0.85 and subsample=1.0.

Technical rationale: Comparable Store+Department models are Linear Regression, Random Forest and tuned LightGBM. Prophet and LSTM are aggregate-level exploratory models and are not mixed into the primary ranking.

7. Prediction & Post-Processing

The best model is selected programmatically by minimum WMAE. Validation forecasts are clipped at zero. For test data, the notebook performs recursive walk-forward forecasting: build lag/rolling features, predict one week, append the prediction to history, then continue.

Technical rationale: The test process covers 39 weeks and produces 115,064 forecasts in Kaggle format: Id = Store_Dept_Date and Weekly_Sales.

8. Evaluation & Robustness

WMAE is the primary metric, with holiday observations weighted 5×. Random Forest records MAE 1,371.55, RMSE 2,875.41, WMAE 1,400.53, MAPE 59.40% and R^2 0.98553. Robustness is reported for holiday status, store type and markdown level.

Technical rationale: Holiday WMAE is 1,502.14 versus 1,361.52 for non-holidays; Type A WMAE is 1,739.93 versus 1,073.77 for Type B; high-markdown WMAE is 1,471.81.

Residual note: the notebook visualizes residuals versus predictions, residual distribution and residuals over time, and reports store/department error rankings. It does not implement a formal statistical normality test.

9. Explainability

SHAP TreeExplainer is used for the winning tree model and provides global and local explanations. The top drivers are Sales_lag_52, Sales_lag_1, Sales_roll_mean_4, Sales_roll_mean_8 and Sales_roll_mean_12. Three local SHAP examples and a LIME local cross-check are generated.

Technical rationale: The computed feature ranking is translated into inventory-planning guidance rather than relying on static assumptions.

10. Reproducibility & Production Shape

The pipeline is naturally suited to weekly batch forecasting: ingest new data, rebuild features, retrain on expanding history, monitor segment-level WMAE and generate the next horizon. The recursive test loop already mirrors this production shape.

Technical rationale: Production hardening should add model/version tracking, automated data-quality checks, drift alerts and robust cold-start fallbacks.

11. Reproduction Checklist

Stage	Checkpoint
Data	Place stores.csv, train.csv, test.csv and features.csv where the notebook expects them.
Cleaning	Confirm markdown fill, store-wise macro-variable imputation and negative-sales audit.
Features	Recreate calendar, holiday, lag and rolling features in Store+Dept+Date order.
Validation	Use chronological hold-out; do not replace it with random splitting.
Tuning	Run TimeSeriesSplit for LightGBM and retain fold scores.
Selection	Rank comparable Store+Department models by WMAE.
Explainability	Run SHAP global/local explanations and the LIME cross-check.
Forecast	Run recursive test forecasting and verify 115,064 rows.
Submission	Verify reports/submission.csv contains Id and Weekly_Sales.

This document describes the implemented v2 notebook and explicitly calls out any requirement that is visually/partially covered rather than claiming undocumented work was completed.